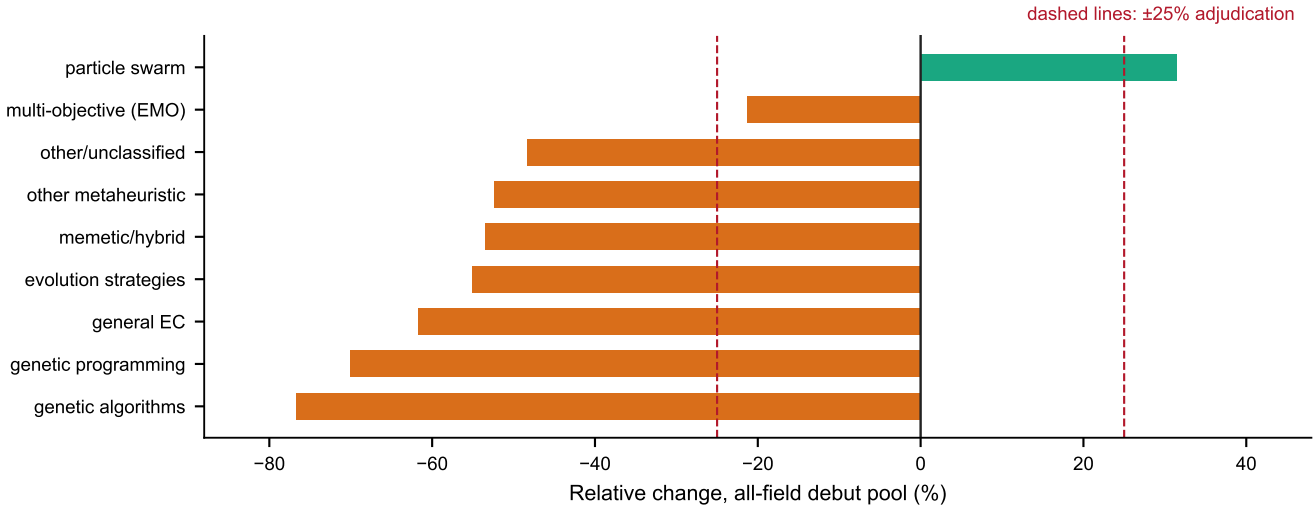
a Eligible subfamilies (n≥200 both periods)



b Subfamily audit table, all-field debut pool

subfamily	n early	n modern	relative change	BH p	eligible
particle swarm	457	1,773	+31.5%	1.00	yes
multi-objective (EMO)	839	1,948	-21.3%	<1e-6	yes
other/unclassified	472	720	-48.3%	<1e-6	yes
other metaheuristic	509	714	-52.4%	<1e-6	yes
memetic/hybrid	253	347	-53.5%	<1e-6	yes
evolution strategies	235	311	-55.1%	<1e-6	yes
general EC	1,309	1,479	-61.7%	<1e-6	yes
genetic programming	596	526	-70.1%	<1e-6	yes
genetic algorithms	1,725	1,185	-76.7%	<1e-6	yes
other swarm/bio	119	1,697	+383.3%	—(n<200)	no
differential evolution	143	868	+106.3%	—(n<200)	no
neuroevolution	56	229	+38.7%	—(n<200)	no
generic heuristic	162	540	+13.3%	—(n<200)	no
hyper-heuristic	22	65	+0.0%	—(n<200)	no
ant colony	195	379	-34.3%	—(n<200)	no
estimation-of-distribution	103	105	-64.9%	—(n<200)	no
genetic fuzzy	31	21	-76.5%	—(n<200)	no
learning classifier systems	139	39	-90.9%	—(n<200)	no

Pool: all-field first-paper authors, 2003–07 vs 2019–22; eligible = n≥200 in both periods. 'no' rows are descriptive (no test).